

SMA 2304: ORDINARY DIFFERENTIAL EQUATIONS I

TUTORIAL ONE

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- a. The number $N(t)$ of individuals in JKUAT who have heard a rumor at time t is given by

$N(t) = ce^{-2e^{-0.5t}}$ obtain a differential equation for the rate of the propagation of the rumor.

- b. Show that the following equations are exact hence determine their solutions

i) $(3x^2 + 4xy)dx + (2x^2 + 2y)dy = 0$

ii) $(2x \cos y + 3x^2 y)dx + (x^3 - x^2 \sin y - y)dy = 0$

iii) $(3x + 2y)dx + (2x + y)dy = 0$

iv) $(3x^2 y^2 + 4x^3 y^2)dx + (2x^3 y + 3x^4 y^2)dy = 0$

v) $\left(\frac{3-y}{x^2}\right)dx + \left(\frac{y^2 - 2x}{xy^2}\right)dy = 0$

- c. In each of the following equations determine the constants A such that the equation is exact hence solve the resulting equation.

i) $(x^2 + 3xy)dx + (Ax^2 + 4y)dy = 0$

ii) $\left(\frac{1}{x^2} + \frac{1}{y^2}\right)dx + \left(\frac{Ax+1}{y^2}\right)dy = 0$

- d. Consider the differential equation

$$(4x + 3y^2)dx + 2xydy = 0$$

- i) Show that this equation is not exact.
ii) Find an integrating factor of form x^n where n is a positive integer.
iii) Multiply the given equation through by the integrating factor found in b and solve the resulting equation.

- e. The number of words N per minute a first year Actuarial student can type using a computer increases with practice. If the rate of change of N is proportional to the difference between N and an upper limit of 100 words, assuming that at the start of the year the student could not type at all and after 10 days of practice the student can type 25 words.

- a. How many words can the student type after 15 days of practice?
b. How many days must a student practice to be able to type 50 words?

NB: (Take note of the concept below and transfer it to your notebooks after the second lecture.

It will assist you to solve part d above)

INTEGRATING FACTORS

Consider the equation $M(x, y)dx + N(x, y)dy = 0$

If $\frac{\partial M(xy)}{\partial y} = \frac{\partial N(xy)}{\partial x}$ then the equation is exact, the solution of which is one parameter family of

solutions. If $\frac{\partial M(xy)}{\partial y} \neq \frac{\partial N(xy)}{\partial x}$ the equation is not exact thus the above method cannot be applied

to solve the equation.

Perhaps this equation can be multiplied by an expression which will transform it to an exact one e.g. consider the equation

$ydx + 2xdy = 0$, this equation is not exact but if multiplied by y it is transformed to an exact one $y^2dx + 2xydy = 0$. Since this equation is integrable y is called the integrating factor.

Definition- if the equation $M(x, y)dx + N(x, y)dy = 0$ is not exact in domain D but the differential equation $\mu(xy)M(xy)dx + \mu(xy)N(xy)dy = 0$ is exact in D then $\mu(xy)$ is called an integrating factor of the given equation.

Example: Consider the differential equation $(3y + 4xy^2)dx + (2x + 3x^2y)dy = 0$.

Now $\frac{\partial M}{\partial y} = 3 + 8xy$ and $\frac{\partial N}{\partial x} = 2 + 6xy$. This equation is not exact. Let $\mu(xy) = x^2y$ then multiply

the given equation by $\mu(xy)$ we have $(3x^2y^2 + 4x^3y^2)dx + (2x^3y + 3x^4y^2)dy$ which is exact since

$$\frac{\partial M}{\partial y}(xy)\mu = 6x^2y + 12x^3y^2$$

$$\frac{\partial N}{\partial x}\mu(xy) = 6x^2y + 12x^3y^2 \text{ hence } \mu(xy) \text{ is the integrating factor of this equation.}$$

End of note.

f. Solve the initial value problems

i) $\left(y + \sqrt{x^2 + y^2}\right)dx - xdy = 0 \quad y(1) = 0$

ii) $(2xy + 3y^2)dx - (2xy + x^2)dy = 0 \quad y(1) = -2$

g. Solve the following equations

i) $\frac{dy}{dx} + \left(\frac{2x+1}{x}\right)y = e^{-2x}$

ii) $(x^2 - 1)\frac{dy}{dx} + 4xy = x$

iii) $x^4\frac{dy}{dx} + 2x^3y = 1$

iv) $\frac{dy}{dx} + y = xy^3$

v) $\frac{dy}{dx} - \frac{y}{x} = -\frac{y^2}{x}$

vi) $\frac{dy}{dx} + \frac{y}{2x} = \frac{x}{y^3}$